

Day 01 Practice Assignments: Python Intro & Control Flow

Objective

Apply variables, operators, loops, and conditional statements to solve introductory algorithmic challenges.

Exercise 1: Leap Year Checker

Write a program that takes a year as input from the user and checks whether it is a leap year or not.

- **Leap Year Criteria:** A year is a leap year if it is divisible by 4, except for century years (ending in 00), which must also be divisible by 400.
 - **Sample Input:** 2024
 - **Sample Output:** 2024 is a Leap Year.
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Exercise 2: Fibonacci Sequence Generator

Write a Python script to print the first N terms of the Fibonacci sequence, where N is provided by the user.

- **Fibonacci sequence:** 0, 1, 1, 2, 3, 5, 8, 13, 21, ...
 - **Sample Input:** $N = 6$
 - **Sample Output:** 0, 1, 1, 2, 3, 5
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Exercise 3: Prime Number Checker

Write a program that checks whether a positive integer entered by the user is a prime number.

- **Logic:** A prime number is a number greater than 1 that has no positive divisors other than 1 and itself.
 - **Sample Input:** 17
 - **Sample Output:** 17 is a prime number.
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Exercise 4: Odd or Even Checker

Write a program that prompts the user for an integer and prints whether it is even or odd.

- **Sample Input:** 7
 - **Sample Output:** 7 is an Odd number.
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Exercise 5: Basic Operator Calculator

Create a program that takes two numbers and a math operator (+, -, *, /) from the user, performs the corresponding calculation, and prints the result.

- **Sample Input:** num1=15, num2=3, operator='/'
 - **Sample Output:** Result: 5.0
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Exercise 6: Sum of N Natural Numbers

Write a script that accepts a positive integer N from the user and calculates the sum of all natural numbers up to N .

- **Formula:** $\sum_{i=1}^N i = \frac{N(N+1)}{2}$
 - **Sample Input:** N = 10
 - **Sample Output:** Sum: 55
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Exercise 7: Multiplication Table Generator

Write a program that takes an integer from the user and prints its multiplication table from 1 to 10.

- **Sample Input:** 5
- **Sample Output:**

```
5 x 1 = 5
5 x 2 = 10
...
5 x 10 = 50
```

Exercise 8: Score to Grade Converter

Write a script that takes a numeric test score from the user (0 to 100) and displays a corresponding letter grade based on the following scale:

- 90-100: A
- 80-89: B
- 70-79: C
- 60-69: D
- Below 60: F